[System Programming]

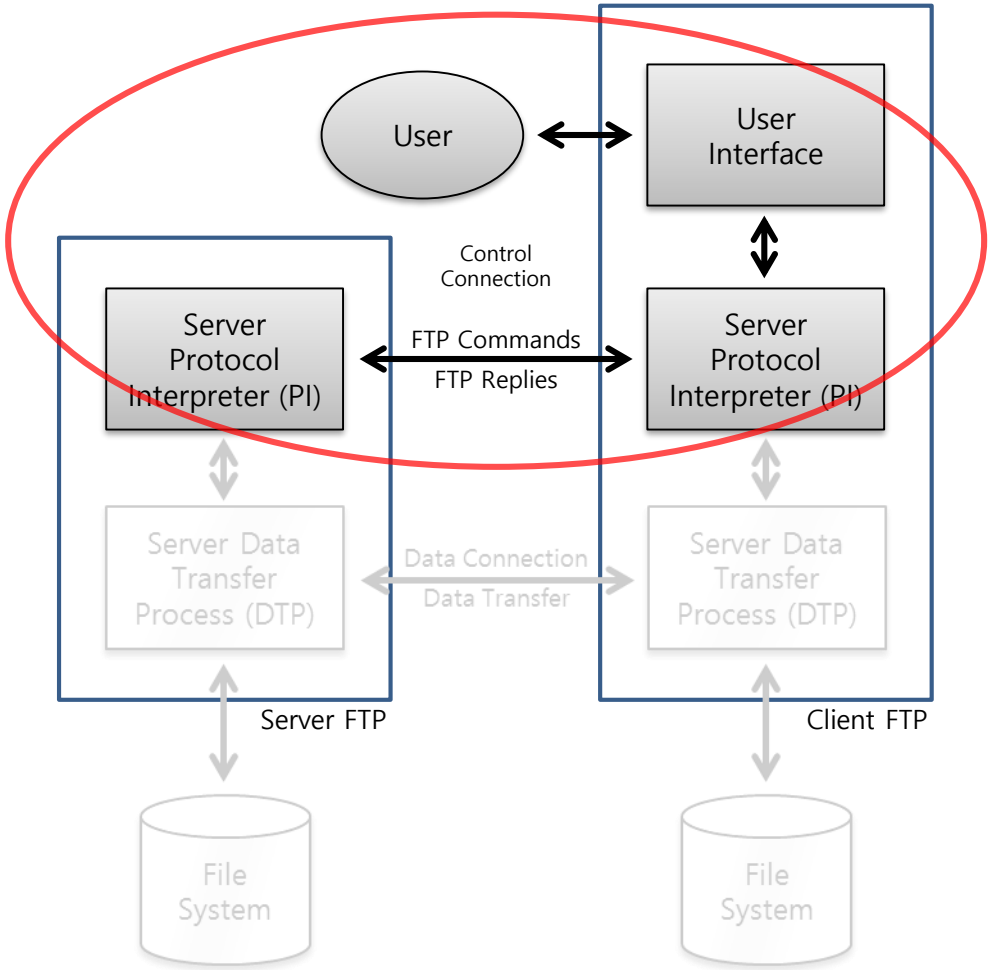
FTP 1-3

Healthcare and Artificial Intelligence Lab.

#FTP Server

Assignment #1 (1/6)

- FTP Commands
  - Access Control Commands
    - USER, PASS, ACCT, CWD, CDUP, SMNT, REIN, QUIT
  - Transfer Parameter Commands
    - PORT, PASV, TYPE, STRU, MODE
  - FTP Service Commands
    - RETR, STOR, STOU, APPE, ALLO, REST, RNFR, RNTO, ABOR, DELE, RMD, MKD, PWD, LIST, NLST, SITE, SYST, STAT, HELP, NOOP



Assignment #1 (2/6)

- Implementation two programs
  - Convert user command to FTP commands (cli)
  - Check FTP commands and execute your commands (srv)

- Convert user command to FTP commands
  - cli program's inputs are user command, option and/or pathnames as arguments
    - `$ cli ls -al [path]`
  - cli program's outputs are FTP command and options and/or pathnames as string
    - `NLST -al [path]`
  - Using write system call for output (standard output)

Assignment #1 (3/6)

- cli program's requirements
  - Check arguments
  - Convert command
    - Separate USER command from arguments
    - Change USER command to FTP command
    - Combine FTP command and optional argument
  - Input the converted command to the output buffer
  - Write the buffer's contents to STANDARD OUTPUT
    - Use write system call
  - Check all inputs for exception handling
    - Command
    - Options
    - non-option arguments

Assignment #1 (4/6)

- Converting
  - ls : NLST ➔ name list
  - ls -a : NLST -a ➔ name list all
  - ls -l : NLST -l ➔ name list long format
  - ls -al : NLST -al ➔ name list all and long format
  - dir : LIST ➔ directory entry list
  - pwd : PWD ➔ print current working directory
  - cd : CWD ➔ change working directory
  - cd .. : CDUP ➔ change directory to ..
  - mkdir: MKD ➔ make directory
  - delete : DELE ➔ remove file
  - rmdir : RMD ➔ remove directory
  - rename: RNFR & RNTO ➔ rename from and to
  - get & put : RETR & STOR ➔ retrieve from and store to
  - quit: QUIT ➔ terminates the command connection

Assignment #1 (5/6)

- Check FTP commands and execute your commands
  - srv program's inputs are FTP command and options and/or pathnames
    - using read system call
    - through STANDARD INPUT
  - srv program's output are your own commands' execution results
    - using write system call
    - through STANDARD OUTPUT

Assignment #1 (6/6)

- srv program's requirements
  - Check STANDARD INPUT
    - use read system call
  - Convert command
    - take advantage of cli program's algorithm
  - Execute your own function
  - Display result
    - use write system call
  - Check the error conditions and results

Requirements (1/6)

- 구현하는 모든 명령어는 FTP 명령어를 기반으로 함.

- 예외처리
  - 알 수 없는 옵션 및 인자 처리
  - 접근 권한 유무 판별 후 처리

- Terminal(shell)에 data를 출력, 받아올 수 있는 library 함수
  - 이는 모두 write(), read() system call로 대체하여 사용

Requirements (2/6)

- ls (NLST) Requirements
  - -a, -l, -al 옵션 구현
  - -a 옵션이 없을 경우 숨김 파일 및 디렉토리가 보여서는 안됨
  - 디렉토리 뒤에만 '/'을 붙여서 파일과 디렉토리를 구분
    - (e.g. ./, ../, this_is_dir/, this_is_file)
    - 리눅스 명령어 $ls -F 참고
  - -l option 없을 때 한 줄에 출력 가능한 파일들을 임의로 정함(본 프로그램은 5개)
  - 출력은 ASCII값 기준으로 좌→우로 1-row씩 오름차순 정렬
  - ls [directory name]은 모든 옵션(-a, -l, -al)포함
  - ls [File_name]은 옵션(-l) 포함
    - +tip (Assignment1-2에서 오류처리 했던 부분 수정)
  - 잘못된 경로, 잘못된 파일 입력, 잘못된 옵션에 대하여 오류 처리
  - 접근권한 없는 파일, 디렉토리에 대하여 오류 처리
  - Wildcard 구현 x(e.g. $ls *)
  - -l option 구현 시 total 구현 x

Requirements (3/6)

- dir (List) Requirements
  - Options & Arguments
    - 출력하고자 하는 파일 등의 이름 또는 경로
  - Behaviors
    - ls -al (NLST -al)와 동일

- pwd (PWD) Requirements
  - Options & Arguments
    - 없음
  - Behaviors
    - Current working directory 출력

Requirements (4/6)

- cd (CWD) / cd .. (CDUP) Requirements
  - Options & Arguments
    - 이동할 디렉토리 이름 또는 경로
    - 디렉토리 이름 또는 경로명이 틀린 경우, 예외처리
    - 디렉토리명 미입력 시 예외처리
  - Behaviors
    - Current working director의 이동

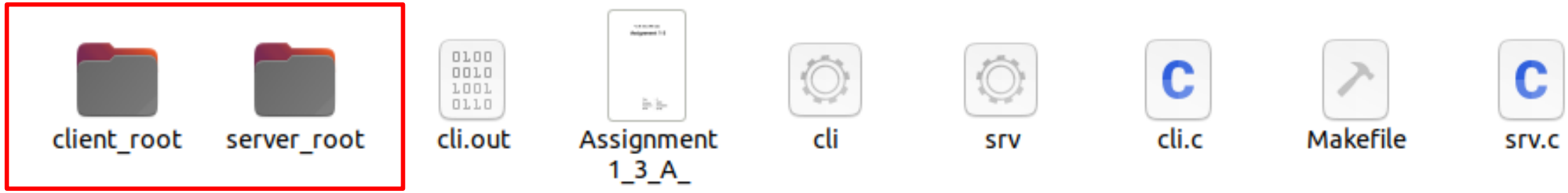
- Mkdir (MKD) Requirements
  - Options & Arguments
    - 생성할 디렉토리 명
    - 복수 가능
    - 이미 존재하는 디렉토리인 경우, 디렉토리 명 미입력 시 예외처리
  - Behaviors
    - 디렉토리 생성

Requirements (5/6)

- delete (DELE) / rmdir (RMD) Requirements
  - Options & Arguments
    - 삭제할 디렉토리 / 파일명
    - 복수 가능
    - 디렉토리 / 파일명이 없는 경우 예외처리
  - Behaviors
    - 디렉토리 / 파일 삭제
- rename (RNFR & RNTO) Requirements
  - Options & Arguments
    - 원본 디렉토리 / 파일명과 변경할 디렉토리 / 파일명
    - 이미 존재하는 디렉토리인 경우, 디렉토리 명 미입력 시 예외처리
  - Behaviors
    - 디렉토리 / 파일 명 변경 및 이동
- quit (QUIT) Requirements
  - Options & Arguments
    - 없음
  - Behaviors
    - QUIT 성공 메시지 출력

Requirements (6/6)

- Before implementation
  - Create ‘client_root’ and ‘server_root’ directories before command execution
  - Prepare the root directory structure required for put and get operations



Requirements (6/6)

- get (RETR) Requirements
  - Options & Arguments
    - ➢ 없음
  - Behaviors
    - ➢ 'server_root' 에 존재하는 파일 불러오기
      - ✓ Client가 get 요청을 Server 로 보낸다.
      - ✓ Server는 'server_root'에 파일 존재 확인
        - » 존재 : 'client_root'에 텍스트 파일을 생성 후, 내용을 write하면서 read byte 출력
        - » 존재x : 에러 문구
- put (STOR) Requirements
  - Options & Arguments
    - ➢ 없음
  - Behaviors
    - ➢ 'client_root'에 존재하는 파일 'server_root'파일로 보내기
      - ✓ Client가 put할 파일 존재 'client_root'내에서 확인
        - » 존재: 텍스트 파일 내용 buffer에 저장 후, Server는 파일 생성 후 write하면서 write byte 출력
        - » 존재x: 에러 문구

Result example-NLST

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -e | tee cli.out | ./srv
Error: invalid option
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls | tee cli.out | ./srv
NLST
Assignment1_3.pdf    Makefile    cli    cli.c    cli.out
client_root/    server_root/    srv    srv.c    test_dir_1/
test_dir_2/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -a | tee cli.out | ./srv
NLST    -a
./    ../    Assignment1_3.pdf    Makefile    cli
cli.c    cli.out    client_root/    server_root/    srv
srv.c    test_dir_1/    test_dir_2/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -l | tee cli.out | ./srv
NLST    -l
-rwxrw-rw-  1 kw2025190614 kw2025190614 960377 Apr 18 04:05 Assignment1_3.pdf
-rw-rw-r--  1 kw2025190614 kw2025190614    60 Apr 18 05:01 Makefile
-rwxrwxr-x  1 kw2025190614 kw2025190614 21112 Apr 17 23:02 cli
-rw-rw-r--  1 kw2025190614 kw2025190614  8745 Apr 18 04:15 cli.c
-rw-rw-r--  1 kw2025190614 kw2025190614     7 Mar 29 05:20 cli.out
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 client_root/
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 server_root/
-rwxrwxr-x  1 kw2025190614 kw2025190614 26776 Apr 18 03:59 srv
-rw-rw-r--  1 kw2025190614 kw2025190614 22590 Apr 18 05:01 srv.c
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_1/
d---------  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_2/
```

Result example-NLST(Cont'd)

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -al | tee cli.out | ./srv
NLST    -al
drwxrwxr-x  6 kw2025190614 kw2025190614  4096 Mar 29 05:19 ./
drwxrwxr-x  3 kw2025190614 kw2025190614  4096 Mar 29 05:13 ../
-rwxrw-rw-  1 kw2025190614 kw2025190614 960377 Apr 18 04:05 Assignment1_3.pdf
-rw-rw-r--  1 kw2025190614 kw2025190614    60 Apr 18 05:01 Makefile
-rwxrwxr-x  1 kw2025190614 kw2025190614 21112 Apr 17 23:02 cli
-rw-rw-r--  1 kw2025190614 kw2025190614  8745 Apr 18 04:15 cli.c
-rw-rw-r--  1 kw2025190614 kw2025190614     8 Mar 29 05:20 cli.out
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 client_root/
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 server_root/
-rwxrwxr-x  1 kw2025190614 kw2025190614 26776 Apr 18 03:59 srv
-rw-rw-r--  1 kw2025190614 kw2025190614 22590 Apr 18 05:01 srv.c
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_1/
d---------  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_2/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -al ./not_exist_path | tee cli.out | ./srv
NLST    -al
Error : No such file or directory
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -al ./test_dir_1 | tee cli.out | ./srv
NLST    -al
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 ./
drwxrwxr-x  6 kw2025190614 kw2025190614  4096 Mar 29 05:19 ../
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -al ./test_dir_2 | tee cli.out | ./srv
NLST    -al
Error: cannot access
```

Result example-NLST(Cont'd)

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -al | tee cli.out | ./srv
NLST    -al
drwxrwxr-x  6 kw2025190614 kw2025190614  4096 Mar 29 05:19 ./
drwxrwxr-x  3 kw2025190614 kw2025190614  4096 Mar 29 05:13 ../
-rwxrw-rw-  1 kw2025190614 kw2025190614 960377 Apr 18 04:05 Assignment1_3.pdf
-rw-rw-r--  1 kw2025190614 kw2025190614    60 Apr 18 05:01 Makefile
-rwxrwxr-x  1 kw2025190614 kw2025190614 21112 Apr 17 23:02 cli
-rw-rw-r--  1 kw2025190614 kw2025190614  8745 Apr 18 04:15 cli.c
-rw-rw-r--  1 kw2025190614 kw2025190614     8 Mar 29 05:20 cli.out
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 client_root/
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 server_root/
-rwxrwxr-x  1 kw2025190614 kw2025190614 26776 Apr 18 03:59 srv
-rw-rw-r--  1 kw2025190614 kw2025190614 22590 Apr 18 05:01 srv.c
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_1/
d---------  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_2/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -al ./not_exist_path | tee cli.out | ./srv
NLST    -al
Error : No such file or directory
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -al ./test_dir_1 | tee cli.out | ./srv
NLST    -al
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 ./
drwxrwxr-x  6 kw2025190614 kw2025190614  4096 Mar 29 05:19 ../
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli ls -al ./test_dir_2 | tee cli.out | ./srv
NLST    -al
Error: cannot access
```

Result example-LIST

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli dir | tee cli.out | ./srv
LIST
drwxrwxr-x  6 kw2025190614 kw2025190614  4096 Mar 29 05:19 ./
drwxrwxr-x  3 kw2025190614 kw2025190614  4096 Mar 29 05:13 ../
-rwxrw-rw-  1 kw2025190614 kw2025190614 960377 Apr 18 04:05 Assignment1_3.pdf
-rw-rw-r--  1 kw2025190614 kw2025190614    60 Apr 18 05:01 Makefile
-rwxrwxr-x  1 kw2025190614 kw2025190614 21112 Apr 17 23:02 cli
-rw-rw-r--  1 kw2025190614 kw2025190614  8745 Apr 18 04:15 cli.c
-rw-rw-r--  1 kw2025190614 kw2025190614     4 Mar 29 05:22 cli.out
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 client_root/
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 server_root/
-rwxrwxr-x  1 kw2025190614 kw2025190614 26776 Apr 18 03:59 srv
-rw-rw-r--  1 kw2025190614 kw2025190614 22590 Apr 18 05:01 srv.c
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_1/
d---------  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_2/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli dir -e | tee cli.out | ./srv
Error: invalid option
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli dir ./not_exist_path | tee cli.out | ./srv
Error : No such file or directory
```

Result example-LIST

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli dir | tee cli.out | ./srv
LIST
drwxrwxr-x  6 kw2025190614 kw2025190614  4096 Mar 29 05:19 ./
drwxrwxr-x  3 kw2025190614 kw2025190614  4096 Mar 29 05:13 ../
-rwxrw-rw-  1 kw2025190614 kw2025190614 960377 Apr 18 04:05 Assignment1_3.pdf
-rw-rw-r--  1 kw2025190614 kw2025190614    60 Apr 18 05:01 Makefile
-rwxrwxr-x  1 kw2025190614 kw2025190614 21112 Apr 17 23:02 cli
-rw-rw-r--  1 kw2025190614 kw2025190614  8745 Apr 18 04:15 cli.c
-rw-rw-r--  1 kw2025190614 kw2025190614     4 Mar 29 05:22 cli.out
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 client_root/
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:10 server_root/
-rwxrwxr-x  1 kw2025190614 kw2025190614 26776 Apr 18 03:59 srv
-rw-rw-r--  1 kw2025190614 kw2025190614 22590 Apr 18 05:01 srv.c
drwxrwxr-x  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_1/
d---------  2 kw2025190614 kw2025190614  4096 Mar 29 05:17 test_dir_2/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli dir -e | tee cli.out | ./srv
Error: invalid option
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli dir ./not_exist_path | tee cli.out | ./srv
Error : No such file or directory
```

Result example-PWD

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli pwd | tee cli.out | ./srv
'/home/kw2025190614/Desktop/FTP1-3/ftp1_3' is current directory
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli pwd -e | tee cli.out | ./srv
Error: invalid option
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli pwd arg | tee cli.out | ./srv
Error: argument is not required
```

Result example-CWD & CDUP

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli cd -e .. | tee cli.out | ./srv
Error: invalid option
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli cd test_dir_1 | tee cli.out | ./srv
CWD   test_dir_1
'/home/kw2025190614/Desktop/FTP1-3/ftp1_3/test_dir_1' is current directory
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli cd .. | tee cli.out | ./srv
CDUP
'/home/kw2025190614/Desktop/FTP1-3' is current directory
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli cd ./not_exist_path | tee cli.out | ./srv
Error: directory not found
```

Result example-MKD

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli mkdir -e new_dir | tee cli.out | ./srv
Error: invalid option
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli mkdir new_dir1 | tee cli.out | ./srv
MKD new_dir1
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli mkdir new_dir2 new_dir3 | tee cli.out | ./srv
MKD new_dir2
MKD new_dir3
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli mkdir new_dir1 | tee cli.out | ./srv
Error: cannot create directory 'new_dir1': File exists
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli mkdir | tee cli.out | ./srv
Error: argument is required
```

Result example-DELE

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ touch test1.dat
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ touch test2.dat
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ touch test3.dat
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ls -al *.dat
-rw-rw-r-- 1 kw2025190614 kw2025190614 0 Mar 29 05:27 test1.dat
-rw-rw-r-- 1 kw2025190614 kw2025190614 0 Mar 29 05:27 test2.dat
-rw-rw-r-- 1 kw2025190614 kw2025190614 0 Mar 29 05:27 test3.dat
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli delete test1.dat | tee cli.out | ./srv
DELE  test1.dat
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ls -al test1.dat
ls: cannot access 'test1.dat': No such file or directory
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli delete test2.dat test3.dat | tee cli.out | ./srv
DELE  test2.dat
DELE  test3.dat
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ls -al *.dat
ls: cannot access '*.dat': No such file or directory
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ touch test.dat
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli delete -e test.dat | tee cli.out | ./srv
Error: invalid option
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli delete not_exist_file test.dat | tee cli.out | ./srv
Error: failed to stat the file or directory
DELE  test.dat
```

Result example-RMD

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ mkdir new_dir_1
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ mkdir new_dir_2
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ mkdir new_dir_3
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli rmdir new_dir_1 | tee cli.out | ./srv
RMD   new_dir_1
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ls -al new_dir_1
ls: cannot access 'new_dir_1': No such file or directory
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli rmdir new_dir_1 | tee cli.out | ./srv
Error: failed to remove 'new_dir_1'
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli rmdir new_dir_2 new_dir_3 | tee cli.out | ./srv
RMD   new_dir_2
RMD   new_dir_3
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ mkdir new_dir
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli rmdir not_exist_dir new_dir | tee cli.out | ./srv
Error: failed to remove 'not_exist_dir'
RMD   new_dir
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli rmdir | tee cli.out | ./srv
Error: argument is required
```

Result example-RNFR & RNTO

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ touch a b
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ls -al a b
-rw-rw-r-- 1 kw2025190614 kw2025190614 0 Mar 29 05:31 a
-rw-rw-r-- 1 kw2025190614 kw2025190614 0 Mar 29 05:31 b
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli rename a b | tee cli.out | ./srv
Error: name to change already exists
%s  kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli rename a c | tee cli.out | ./srv
RNFR    a
RNTO  c
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ls -al a b c
ls: cannot access 'a': No such file or directory
-rw-rw-r-- 1 kw2025190614 kw2025190614 0 Mar 29 05:31 b
-rw-rw-r-- 1 kw2025190614 kw2025190614 0 Mar 29 05:31 c
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli rename | tee cli.out | ./srv
Error: two arguments are required
```

+Tip

- At client : “rename a b” → “RNFR a RNTO b”
- Buffer 이용

Result example - Before RETR & STOR

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ls
Assignment1_3.pdf  cli          cli_updated.c  new_dir3     srv_updated.c
b                  cli.c        Makefile       server_root  test_dir_1
buffer             client_root  new_dir1       srv          test_dir_2
c                  cli.out      new_dir2       srv.c
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ cd client_root/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/client_root$ ls
put_test.txt
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/client_root$ cd ..
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ cd server_root/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/server_root$ ls
get_test.txt
```

Result example - Before RETR & STOR (Cont'd)

- 각 text 파일에 본인 학번 넣기 (감점 요인)
  - server_root : "get_test.txt"

Open ~/Desktop/FTP1-3/ftp1_3/server_root get_test.txt Save

```
Kwangwoon Unviersity
System Programming
2026 1st semester
FTP SERVER - Assignment
Student number : 2025190614
get command
```

  - client_root : "put_test.txt"

Open ~/Desktop/FTP1-3/ftp1_3/client_root put_test.txt Save

```
Kwangwoon Unviersity
System Programming
2026 1st semester
FTP SERVER - Assignment
Student number : 2025190614
put command
```

Result example-RETR

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ cd client_root/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/client_root$ ls
put_test.txt
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/client_root$ cd ..
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli get not_target.txt | ./srv
Error: 'not_target.txt' does not exist in server_root
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli get get_test.txt | ./srv
RETR get_test.txt
OK: 122 bytes copied to client_root
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ cd client_root/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/client_root$ ls
get_test.txt  put_test.txt
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/client_root$ cat get_test.txt
Kwangwoon Unviersity
System Programming
2026 1st semester
FTP SERVER - Assignment
Student number : 2025190614
get command
```

Result example-STOR

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ cd server_root/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/server_root$ ls
get_test.txt
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/server_root$ cd ..
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli put not_target.txt | ./srv
Error: 'not_target.txt' does not exist in client_root
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli put put_test.txt | ./srv
STOR put_test.txt
OK: 122 bytes copied to server_root (expected 122 bytes)
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ cd server_root/
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/server_root$ ls
get_test.txt  put_test.txt
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3/server_root$ cat put_test.txt
Kwangwoon Unviersity
System Programming
2026 1st semester
FTP SERVER - Assignment
Student number : 2025190614
put command
```

Result example-QUIT

```
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli quit | tee cli.out | ./srv
Quit success
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli quit -e | tee cli.out | ./srv
Error: invalid option
kw2025190614@ubuntu:~/Desktop/FTP1-3/ftp1_3$ ./cli quit arg | tee cli.out | ./srv
Error: argument is not required
```

Report Requirements

- Ubuntu 20.04.6 Desktop 64bits 환경에서 채점
- Copy 발견 시 0점 처리
- 보고서 구성
  - 보고서 표지
    - ➢ 수업 명, 과제 이름, 담당 교수님, 학번, 이름 필히 명시
      - ✓ 과제 이름 → Assignment 1-3
  - 과제 내용
    - ➢ Introduction
      - ✓ 과제 소개 - 4줄 이상(background 제외) 작성
    - ➢ 결과 화면
      - ✓ 수행한 내용을 캡처 및 설명
    - ➢ 고찰
      - ✓ 과제를 수행하면서 느낀점 작성
    - ➢ Reference
      - ✓ 과제를 수행하면서 참고한 내용을 구체적으로 기록
      - ✓ 강의자료만 이용한 경우 생략 가능

Report Requirements

- Softcopy Upload
  - 제출 파일
    - ➢ 보고서 + 소스파일 하나의 압축 파일로 압축하여 제출(tar.gz)
    - ➢ 보고서(.pdf. 파일 변환)
    - ➢ 소스코드(**Comment 반드시 포함**)
      - ✓ cli.c, srv.c
      - ✓ Makefile
      - ✓ **실행파일 명: cli, srv**
      - ✓ **소스 코드, 실행파일명 다르게 작성 시 감점**
  - Tar 압축 및 해제 방법
    - ➢ 압축 시 ➔ tar –zcvf [압축 파일명].tar.gz [폴더 명]
    - ➢ 해제 시 ➔ tar –zxvf 파일명.tar.gz
  - 보고서 및 압축 파일 명 양식
  - **Assignment1_3 _*수강분류코드_ 학번_ 이름* 으로 작성**
    - ➢ **예시–이론2(금12)만 수강하는 학생인 경우**
      - ✓ **보고서 Assignment1_3_B_2026123456_곽철이.pdf**
      - ✓ **압축 파일 명: Assignment1_3_B_2026123456_곽철이.tar.gz**
    - ➢ **예시–이론2, 실습2를 수강하는 학생인 경우**
      - ✓ **보고서 Assignment1_3_D_2026123456_곽철이.pdf**
      - ✓ **압축 파일 명: Assignment1_3_D_2026123456_곽철이.tar.gz**

수강 요일	이론1 월6수5	이론2 금12	실습1 목3,4	실습2 금3,4	실습3 수3,4
수강분류 코드	A	B	C	D	E

Report Requirements

- 실습 수업을 수강하는 학생인 경우
  - 실습 과목에 과제를 제출
  - 이론 과목에 간단한 .txt 파일로 제출

실습수업때제출했습니다.	2026-03-03 오후 6:44	텍스트 문서	0KB

    - ➢ 이론 과목에 .txt 파일 미 제출 시 감점
    - ➢ .tar.gz 파일로 제출 하지 않을 시 감점

- 과제 제출
  - **KLAS - 강의 과제 제출**
  - **2026년 4월 18일 토요일 23:59까지 제출**
    - ➢ **딜레이 받지 않음**
      - ✓ 제출 마감 시간 내 미제출 시 해당 과제 0점 처리
      - ✓ 교내 서버 문제 발생 시, 메일로 과제 제출 허용(제출 기한 내)

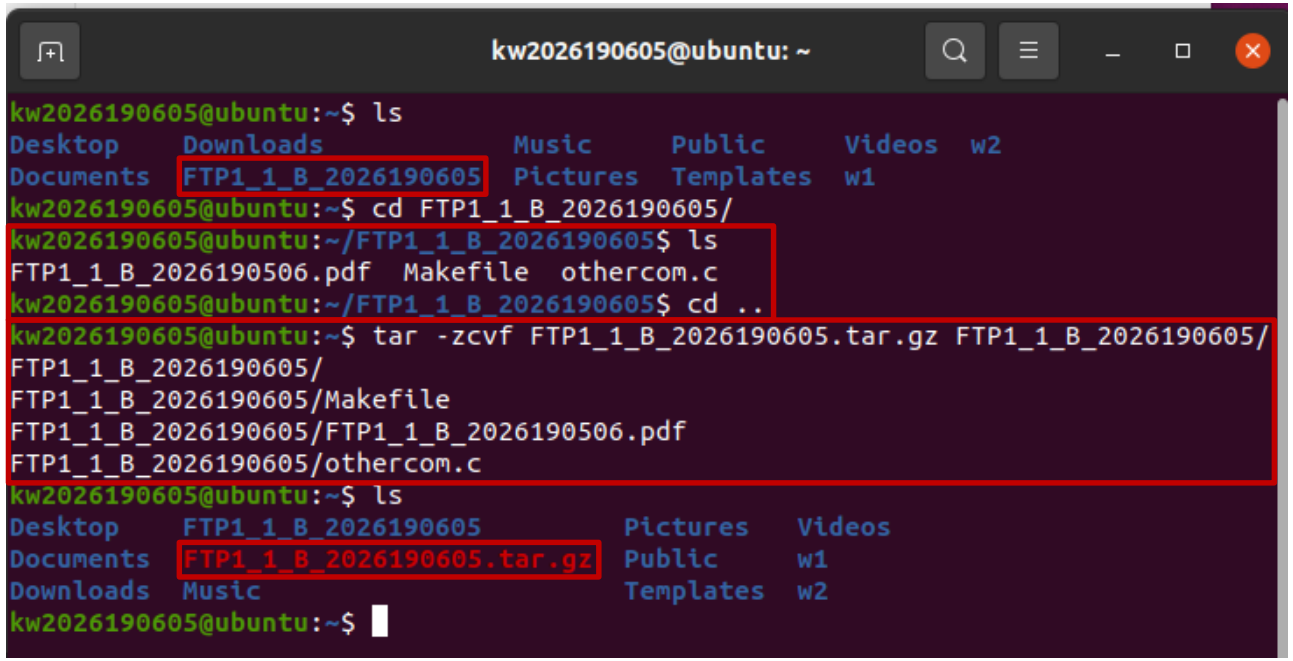
[System Programming]

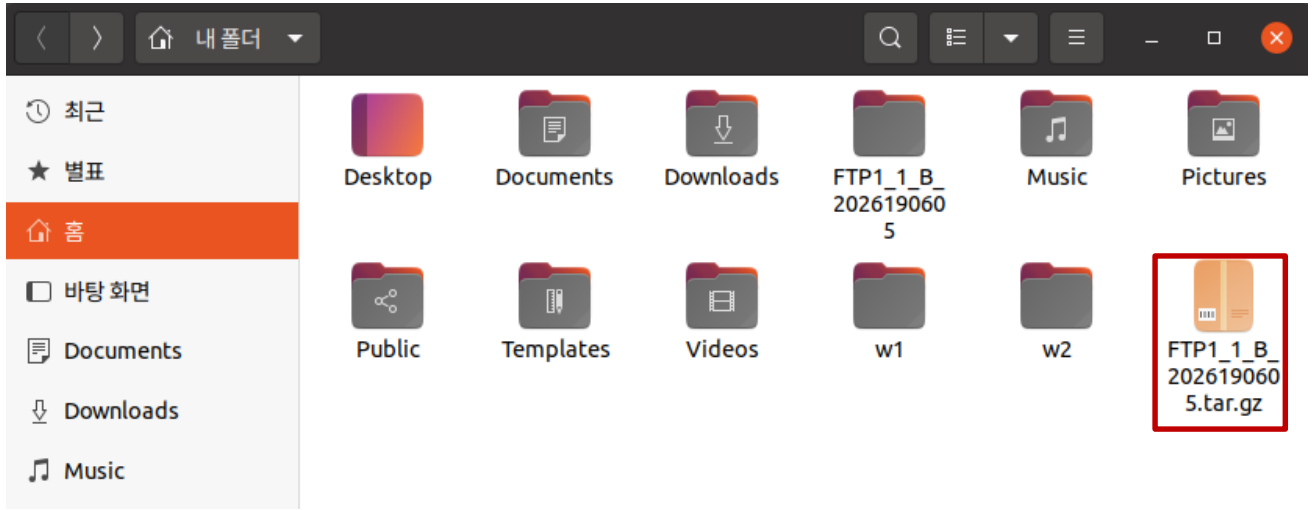
Appendix

Healthcare and Artificial Intelligence Lab.

#FTP Server

Appendix A. tar.gz compression





Appendix B. Comment 작성 요령 (1/3)

- File Head Comment

```
////////////////////////////////////////////////////////////////////////////////
// File Name        : Main.c                                                //
// Date             : 2026/03/01                                            //
// OS          : Ubuntu 20.04.6 LTS 64bits                                  //
// Author      : Gwak Cheol-i                                               //
// Student ID       : 2026123456                                            //
// ------------------------------------------------------------------------ //
// Title       : System Programming Assignment #1-1 ( ftp server )          //
// Description :   ...                                                      //
////////////////////////////////////////////////////////////////////////////////
```

Appendix B. Comment 작성 요령 (2/3)

- Function Head Comment

```
////////////////////////////////////////////////////////////////////////////////
// InsertNode                                                                 //
// ========================================================================== //
// Input: Node* -> Insert Node,                                               //
//        Node* -> Column node before insert node                              //
//        Node* -> Row node before insert node                                //
//        (Input parameter Description)                                       //
// Output: int  -1 success                                                    //
//              0 fail                                                        //
//        (Out parameter Description)                                         //
// Purpose: Inserting node                                                    //
////////////////////////////////////////////////////////////////////////////////
```

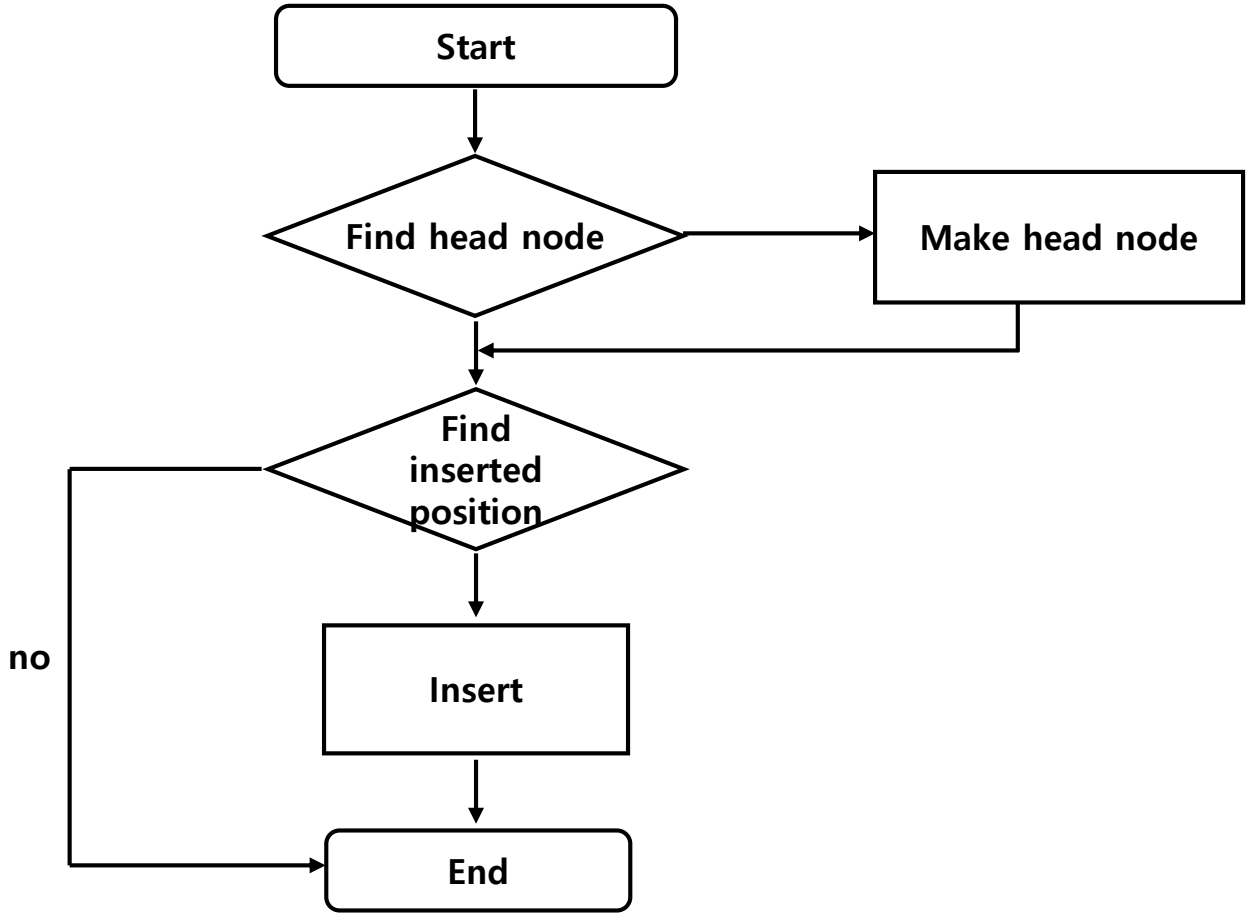
Appendix B. Comment 작성 요령 (3/3)

- In-line Comment

```
//////////////////////////////// Row insert ////////////////////////////////
     if( pRowPos->pNextRow != pRowPos ) {
     pTemp->pNextRow = pRowPos->pNextRow;   // pTemp set next row
     if( !( pRowPos->pNextRow->bHead ) ){
         pRowPos->pNextRow->NodeItem.pPrevRow = pTemp;
     } // end of if
    } // end of if
    else {
     pTemp->pNextRow = pRowPos;    // pTemp set next row
    } // end of else
    pTemp->NodeItem.pPrevRow = pRowPos; // pTemp set previous row
    pRowPos->pNextRow = pTemp;
//////////////////////////////// End of row insert ////////////////////////////////
```

Appendix C. 보고서 작성 요령 (1/2)

- Algorithm - Flow Chart (Each function)
  - E.g.



Appendix C. 보고서 작성 요령 (2/2)

- Algorithm - Pseudo Code

```
FixHeap(Node *root, Key k)
{
    Node vacant, largerChild;
    vacant = root;
    while( vacant is not leaf ) {
        largerChild = the child of vacant with the larger key;
        if( k < largerChild's Key ) {
            copy lagerChild's key to vacant;
            vacant = largerChild;
        }
        else exit loop;
    }
}
```

Thank you for listening

Q & A